



JavaScript Expressions

 WIDGET API VERSION 1.4.0

ECMAScript expressions in meta parameters enable dynamic, context-aware iCUE widget configurations evaluated at runtime.

Overview

JavaScript expressions enable widgets to:

- Generate dynamic option lists based on system capabilities
- Calculate default values from current system state
- Create conditional parameter configurations
- Provide localized content and labels
- Integrate with external data sources and APIs

Basic Syntax

ECMAScript expressions are supported in any `data-*` attribute except `data-type`:

```
<!-- Static value -->
<meta
  name="x-icue-property"
  content="brightness"
  data-label="'Brightness'"
  data-type="slider"
  data-default="50"
  data-min="0"
  data-max="100"
  data-step="1"
/>
<!-- Dynamic expression -->
```

```
<meta
  name="x-icue-property"
  content="brightness"
  data-label="tr('Brightness')"
  data-type="slider"
  data-default="SystemUtils.getSystemVolume()"
  data-min="0"
  data-max="100"
  data-step="1"
/>
```

Module Integration

Declaring Modules

Declare JavaScript modules using the `modules` tag in the widget's manifest:

Manifest:

```
"modules": [
  "modules/SystemUtils.mjs",
  "modules/DeviceCapabilities.js"
]
```

Module File Structure

File: `modules/SystemUtils.mjs`

```
export function getCurrentTimezone() {
  return Intl.DateTimeFormat().resolvedOptions().timeZone;
}
export function getAvailableTimezones() {
  return ["America/New_York", "Europe/London", "Europe/Kyiv", "Asia/Tokyo",
"Australia/Sydney"];
}
export function getSystemVolume() {
  return 75;
}
export function isLowLightMode() {
  const hour = new Date().getHours();
```

```
return hour < 7 || hour > 20;  
}
```

Using Module Functions

Exported functions are available through the module's filename (without extension):

Manifest:

```
"modules": [  
  "modules/SystemUtils.mjs"  
]
```

HTML:

```
<meta  
  name="x-icue-property"  
  content="timezone"  
  data-label="'Time Zone'"  
  data-type="combobox"  
  data-values="SystemUtils.getAvailableTimezones()"  
  data-default="SystemUtils.getCurrentTimezone()"  
>  
<meta  
  name="x-icue-property"  
  content="nightMode"  
  data-label="'Night Mode'"  
  data-type="switch"  
  data-default="SystemUtils.isLowLightMode()"  
>
```

Module Requirements

- File extensions: `.mjs` or `.js`
- Export format: ES6 module exports (`export function name() {}`)
- Location: Relative paths from the widget's HTML file
- Multiple modules: Supported for different functionality

Plugin Functions

Plugin functions are available in expressions. Only blocking (synchronous) functions are supported. Asynchronous functions cannot be used in meta parameter expressions. Example with Sensors Data Provider:

Manifest:

```
"required_plugins": [  
  "widgetbuilder.sensorsdatapprovider:Sensors:1.0"  
]
```

HTML:

```
<meta  
  name="x-icue-property"  
  content="selectedSensorId"  
  data-label="'Sensor'"  
  data-type="sensors-combobox"  
  data-default="plugins.Sensorsdatapprovider.getDefaultSensorIdBlock('temperature')"  
>  
<!-- Only blocking (synchronous) functions are supported. Asynchronous functions  
cannot be used in meta parameter expressions. -->
```